

Import libraries

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

Load dataset

```
df = pd.read_csv("/content/sample_data/CAR DETAILS FROM CAR DEKHO.csv")
```

Data cleaning

```
df.shape
```

```
(4340, 8)
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 4340 entries, 0 to 4339
Data columns (total 8 columns):
 #   Column          Non-Null Count  Dtype  
---  -
 0   name            4340 non-null   object 
 1   year            4340 non-null   int64  
 2   selling_price   4340 non-null   int64  
 3   km_driven       4340 non-null   int64  
 4   fuel            4340 non-null   object 
 5   seller_type     4340 non-null   object 
 6   transmission    4340 non-null   object 
 7   owner          4340 non-null   object 
dtypes: int64(3), object(5)
memory usage: 271.4+ KB
```

```
df.columns
```

```
Index(['name', 'year', 'selling_price', 'km_driven', 'fuel',
       'seller_type',
       'transmission', 'owner'],
      dtype='object')
```

```
df.describe()
```

	year	selling_price	km_driven
count	4340.000000	4.340000e+03	4340.000000
mean	2013.090783	5.041273e+05	66215.777419
std	4.215344	5.785487e+05	46644.102194
min	1992.000000	2.000000e+04	1.000000
25%	2011.000000	2.087498e+05	35000.000000
50%	2014.000000	3.500000e+05	60000.000000
75%	2016.000000	6.000000e+05	90000.000000
max	2020.000000	8.900000e+06	806599.000000

```
df.head()
```

	name	year	selling_price	km_driven	fuel	seller_type	transmissi
0	Maruti 800 AC	2007	60000	70000	Petrol	Individual	Mar
1	Maruti Wagon R LXI Minor	2007	135000	50000	Petrol	Individual	Mar
2	Hyundai Verna 1.6 SX	2012	600000	100000	Diesel	Individual	Mar
3	Datsun RediGO T Option	2017	250000	46000	Petrol	Individual	Mar
4	Honda Amaze VX i-DTEC	2014	450000	141000	Diesel	Individual	Mar

```
pd.isnull(df).sum()
```

	0
name	0
year	0
selling_price	0
km_driven	0
fuel	0
seller_type	0
transmission	0
owner	0

dtype: int64

```
df.dropna(inplace = True)
```

```
df.drop(['seller_type'],axis = 1,inplace = True)
```

```
df.columns
```

```
Index(['name', 'year', 'selling_price', 'km_driven', 'fuel',
      'transmission',
      'owner'],
      dtype='object')
```

```
df.rename(columns = {'name' : 'model' , 'selling_price' : 'Amount' , 'transmission' : 'Gearbox' , 'owner' : 'owner'})
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 4340 entries, 0 to 4339
Data columns (total 7 columns):
#   Column      Non-Null Count  Dtype  
---  -
0   model       4340 non-null   object  
1   year        4340 non-null   int64   
2   Amount      4340 non-null   int64   
3   Mileage     4340 non-null   int64   
4   fuel        4340 non-null   object  
5   Gearbox     4340 non-null   object  
6   owner       4340 non-null   object  
dtypes: int64(3), object(4)
memory usage: 237.5+ KB
```

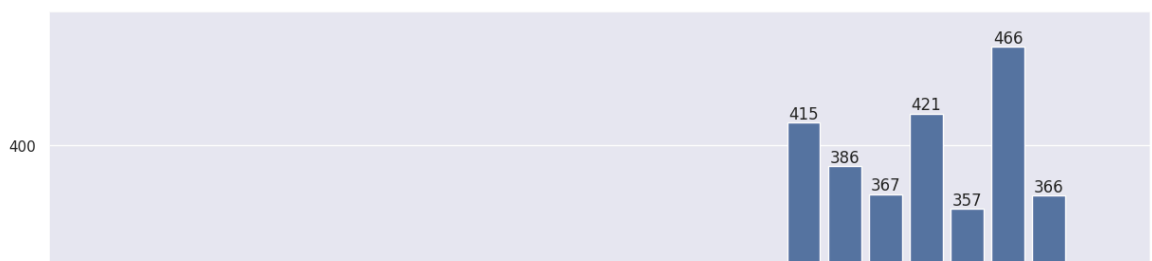
```
df.describe()
```

	year	Amount	Mileage
count	4340.000000	4.340000e+03	4340.000000
mean	2013.090783	5.041273e+05	66215.777419
std	4.215344	5.785487e+05	46644.102194
min	1992.000000	2.000000e+04	1.000000
25%	2011.000000	2.087498e+05	35000.000000
50%	2014.000000	3.500000e+05	60000.000000
75%	2016.000000	6.000000e+05	90000.000000
max	2020.000000	8.900000e+06	806599.000000

Exploratory data analysis

Yearwise

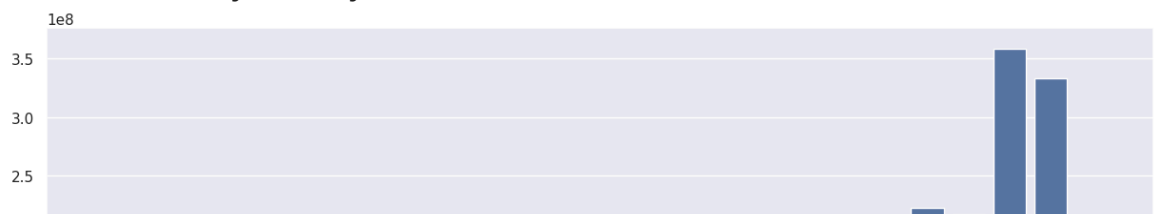
```
sns.set(rc = {'figure.figsize' : (15,10)})
ax = sns.countplot(x = 'year',data = df)
for bars in ax.containers:
    ax.bar_label(bars)
```



```
sales_year = df.groupby(['year'],as_index = False)['Amount'].sum().sort_valu
```

```
sns.set(rc ={'figure.figsize': (15,6)})
sns.barplot(x = 'year',y = 'Amount',data = sales_year)
```

<Axes: xlabel='year', ylabel='Amount'>



Fuelwise

```
sns.set(rc = {'figure.figsize' : (15,10)})
ax = sns.countplot(x = 'fuel',data = df)
```

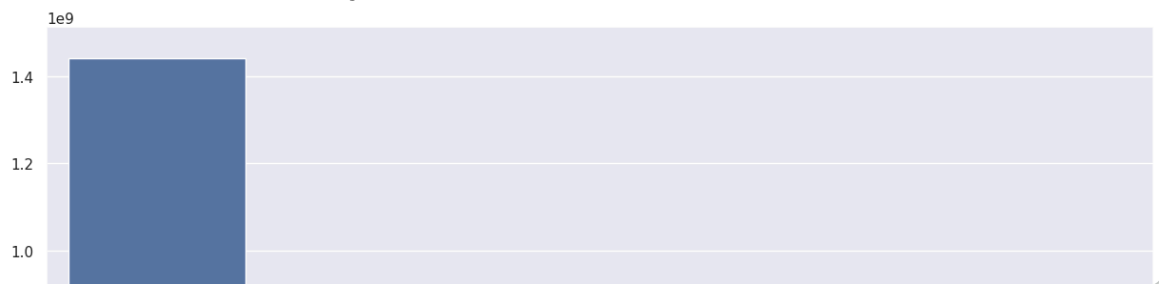
```
for bars in ax.containers:
    ax.bar_label(bars)
```



```
sales_year = df.groupby(['fuel'],as_index = False)['Amount'].sum().sort_valu
```

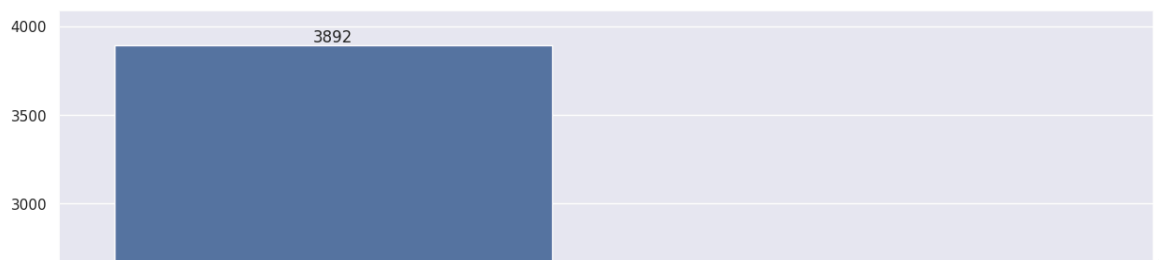
```
sns.set(rc ={'figure.figsize': (15,9)})
sns.barplot(x = 'fuel',y = 'Amount',data = sales_year)
```

<Axes: xlabel='fuel', ylabel='Amount'>



Gearbox

```
sns.set(rc = {'figure.figsize' : (15,10)})
ax = sns.countplot(x = 'Gearbox',data = df)
for bars in ax.containers:
    ax.bar_label(bars)
```



```
sales_year = df.groupby(['Gearbox'],as_index = False)['Amount'].sum().sort_v
```

```
sns.set(rc ={'figure.figsize': (15,9)})
sns.barplot(x = 'Gearbox',y = 'Amount',data = sales_year)
```

```
<Axes: xlabel='Gearbox', ylabel='Amount'>
```



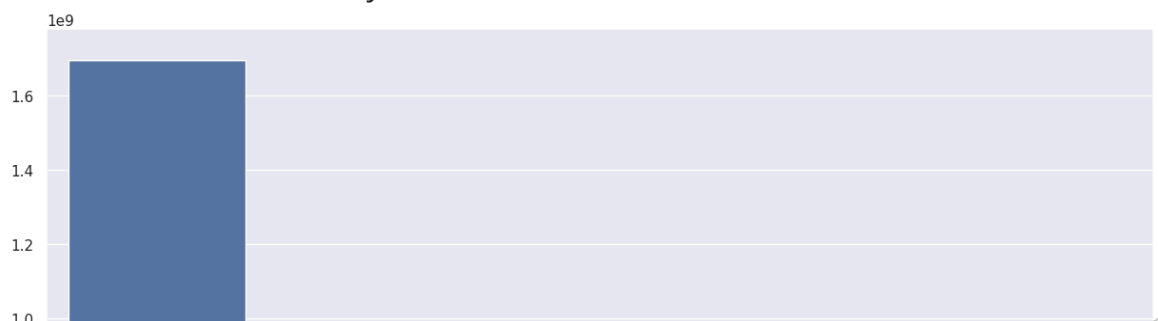
```
sns.set(rc = {'figure.figsize' : (15,10)})  
ax = sns.countplot(x = 'owner',data = df)  
for bars in ax.containers:  
    ax.bar_label(bars)
```



```
sales_year = df.groupby(['owner'],as_index = False)['Amount'].sum().sort_val
```

```
sns.set(rc ={'figure.figsize': (15,9)})  
sns.barplot(x = 'owner',y = 'Amount',data = sales_year)
```

```
<Axes: xlabel='owner', ylabel='Amount'>
```



Final Conclusion

The analysis shows that 2018 had the highest number of car sales, while 2017 generated the highest total selling amount. Diesel cars, manual transmission vehicles, and first-owner cars dominated the market by achieving both the highest sales volume and the highest total selling value.

Advice

Focus on stocking diesel, manual transmission, and first-owner cars, while using the successful sales strategies from 2017–2018 to improve future sales and profitability.